

Argyrios Gerogiannis

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Research Interests

Reinforcement Learning (RL), Reasoning in LLMs and Multimodal Models, Post-Training Methods, Transformer Theory

Education

PhD in Electrical and Computer Engineering

University of Illinois at Urbana-Champaign (UIUC), USA

Cumulative GPA: 4.00/4.00

Aug. 2022 - Jan. 2027 (Expected)

Diploma (BSc+MSc) in Electrical and Computer Engineering (Rank: 1/305)

University of Patras (UoP), Greece

Cumulative GPA: 9.69/10.00

Sep. 2017 - Aug. 2022

Work Experience

Uber Technologies, Inc: Machine Learning Engineer Intern

Summer 2026 / Sunnyvale, CA

- Delivered **two production-facing workstreams** within Uber Risk, each improving an existing production pipeline.
- Developed a transfer-learning approach for fraud detection, improving over production by **44% on Rides** and **10% on Eats**.
- Designed an RL framework for fraud treatment selection, yielding a projected **\$10M+ improvement** over production.

Netradyne, Inc: AI Research Intern

Summer 2025 / San Diego, CA

- Developed a novel end-to-end autonomous driving **vision-encoder RL** stack with **DINOv2**-based vision transformers.
- Achieved **state-of-the-art performance** with **10%** gain over strong baselines on standard driving benchmarks.
- Scaled training via a **distributed** multi-simulator, parallel pipeline across **multi-GPU AWS clusters**, enabling **20x speedup**.

Selected Research Experience

Multimodal Model Post-Training

2026 - Present

- Developed a **low-cost** retroactive tree-based RL method **without online branching** for efficient post-training.
- Improved over state-of-the-art GRPO baselines in Speech LLMs, with up to **39%** performance gains, with **95%** lower training cost.
- Validated across **4 models variants**: Granite 4.0 1B Speech, Granite Speech 3.3 2B & 8B, Qwen-2 Audio 7B.

Large Language Model Reasoning

2025 - Present

- Developed an **MCTS-style framework** for optimal breadth–depth allocation in LLM reasoning under compute budgets.
- Improved accuracy by **+18.8% pass@N** via intrinsic depth refinement on aligned models vs. standard i.i.d. sampling.
- Validated across **8 model variants**: OLMo-3 7B, Qwen-3 4B (Base, Instruct, RLVR) and Nemotron-3 30B (Base, Instruct).

Non-Stationary Reinforcement Learning

2023 - Present

- Derived novel **minimax guarantees** for non-stationary RL and built the first algorithm with matching **optimal bounds**.
- Designed the first **practical black-box** for non-stationary bandits, with **improved order-optimal** performance bounds.
- Attained up to **8x** performance gains and up to **50%** lower compute vs. state-of-the-art baselines in real-world benchmarks.

Publications

DAL: A Practical Prior-Free Black-Box Framework for Piecewise Stationary Bandits
Argyrios Gerogiannis, Yu-Han Huang, Subhonmesh Bose, Venugopal V. Veeravalli
International Conference on Machine Learning - ICML 2026



LEAF: Growing Trees Without Branching for Speech-Aware Large Language Model Post-Training
Argyrios Gerogiannis, Yekaterina Yegorova, Mark Hasegawa-Johnson, Venugopal V. Veeravalli
Conference on Empirical Methods in Natural Language Processing - EMNLP 2026



Towards Context-Based Retrieval in Associative Memories
Moulik Choraria*, **Argyrios Gerogiannis***, Vidhata Jayaraman, Ankur Mani, Lav R. Varshney
ICLR 2026 New Frontiers in Associative Memories Workshop (Oral Presentation)

*Equal contribution



Your Model Diversity, Not Method, Determines Reasoning Strategy
M Choraria, **A Gerogiannis**, A Das, S Chakraborty, B Kapusuzoglu, C-H Lee, K Balasubramaniam, S-X Zhang, S Sahu
ICLR 2026 Workshop on Logical Reasoning of Large Language Models



Learning Where to Look: UCB-Driven Controlled Sensing for Quickest Change Detection
Yu-Han Huang, **Argyrios Gerogiannis**, Subhonmesh Bose, Venugopal V. Veeravalli
IEEE Conference on Decision and Control - CDC 2026 (Oral Presentation)



Detection Augmented Bandit Procedures for Piecewise Stationary MABs: A Modular Approach
Yu-Han Huang, **Argyrios Gerogiannis**, Subhonmesh Bose, Venugopal V. Veeravalli 

IEEE Transactions on Information Theory - TIT 2026

Track-MDP: Reinforcement Learning for Target Tracking with Controlled Sensing (Extended Version) 

Adarsh M. Subramaniam, **Argyrios Gerogiannis**, James Z. Hare, Venugopal V. Veeravalli

IEEE Transactions on Signal Processing - TSP 2026

Is Prior-Free Black-Box Non-Stationary Reinforcement Learning Feasible? 

Argyrios Gerogiannis, Yu-Han Huang, Venugopal V. Veeravalli

International Conference on Artificial Intelligence and Statistics - AISTATS 2025

Track-MDP++: RL for Target Tracking with General Controlled Sensing Models 

Adarsh M. Subramaniam, **Argyrios Gerogiannis**, James Z. Hare, Venugopal V. Veeravalli

IEEE Military Communications Conference - MILCOM 2025

Track-MDP: Reinforcement Learning for Target Tracking with Controlled Sensing 

Adarsh M. Subramaniam, **Argyrios Gerogiannis**, James Z. Hare, Venugopal V. Veeravalli

International Conference on Acoustics, Speech, and Signal Processing - ICASSP 2025

Manuscripts Under Review

SALSA: Speech Aware LLM Adaptation via Learned Steering Activation Vectors 

Yekaterina Yegorova, **Argyrios Gerogiannis**, Haolong Zheng, Julia Hockenmaier, Chang D. Yoo, Mark Hasegawa-Johnson

Under Submission, 2026

Know Thy Reasoner: Not All Language Models Explore Alike 

Moulik Choraria, **Argyrios Gerogiannis**, Anirban Das, Supriyo Chakraborty, Sourya Basu, Sambit Sahu, Lav R. Varshney

Under Submission, 2026

Context-Gated Associative Retrieval: From Theory to Transformers 

Moulik Choraria*, **Argyrios Gerogiannis***, Vidhata Jayaraman, Ankur Mani, Lav R. Varshney

*Equal contribution

Under Submission, 2026

DARLING: Detection Augmented Reinforcement Learning with Non-Stationary Guarantees 

Argyrios Gerogiannis, Yu-Han Huang, Venugopal V. Veeravalli

Under Submission, 2026

Skills

Programming	Python, C++, C, MATLAB
Frameworks	PyTorch, transformers, openrlhf, vLLM, RLlib, ray, TensorFlow, NumPy, pandas, scikit-learn, Hugging Face
Development	Git, Docker, AWS, Slurm, Weights&Biases, MLflow, comet
Languages	Greek (native), English (fluent), Spanish (basic)

Services & Teaching

Reviewer NeurIPS (2026), ICML (2026), AAAI (2027), TMLR (2026), ISIT (2024-2026)

Teaching Assistant UIUC: ECE 561 (2026), ECE 445 (2026), ECE 490 (2025), ECE 313 (2025), ECE 598GM (2023)

Honors & Awards

Gold Reviewer Award

Awarded to top reviewers for ICML 2026; included complimentary registration valued at \$1,030.

ICML, USA

Dilip and Sandhya Sarwate Graduate Fellowship

Awarded to outstanding graduate students.

UIUC, USA

University Student Excellence Award

Awarded to the student with the highest GPA among all graduates of the ECE department.

UoP, Greece

Student Excellence Award

Awarded to the student with the highest GPA among all graduates of each department.

IKY, Greece

Freshman Excellence Award

Awarded to the top 3 freshman students with the highest grade in the entrance exams of the ECE department.

UoP, Greece